

# Assignment: HR AI Chatbot Project

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## 1. Business Requirements Document (BRD)

### a. Problem Statement

At present, the HR team spends a significant amount of time responding to repetitive employee questions. These questions are usually related to common topics such as leave balance, reimbursement procedures, and basic HR policies.

Since most of these queries are handled manually through emails or phone calls, it creates a few ongoing issues. Employees often wait one or two days just to get answers to simple questions. At the same time, HR team members end up spending a large part of their day answering the same queries instead of focusing on more important work like hiring, employee engagement, or process improvements.

Another challenge is inconsistency. Depending on who responds, employees may sometimes receive slightly different answers for the same policy, which leads to confusion and follow-ups.

To solve this, the business needs a reliable digital solution that can provide quick and consistent answers to common HR-related questions, without involving HR staff every single time

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### b. Goals & Objectives

The primary goal of this initiative is to introduce an AI-based HR chatbot that can act as the first level of support for employees.

The key objectives are:

- **Reduce HR workload:** Automate a majority of frequently asked questions so HR teams can focus on strategic tasks. The initial target is to handle around 70% of common queries.
  - **Provide 24/7 access:** Employees should be able to get answers even outside regular working hours.
  - **Improve response consistency:** Ensure that all employees receive the same, approved information for HR policies.
  - **Enhance employee experience:** Make it easier and faster for employees to find HR-related information without delays.
  - **Minimize incorrect responses:** The chatbot will rely only on official company policy documents to reduce chances of incorrect or misleading answers.
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### c. Stakeholders

The following stakeholders are involved in this project:

- **HR Managers:** Own and maintain HR policies. They will validate chatbot responses and ensure information remains accurate.
- **Employees:** End users of the chatbot who need quick and clear answers to HR questions.

- **IT / Technical Team:** Responsible for system setup, security, integrations, and ongoing technical support.
- **Finance Team:** Ensures reimbursement and payroll-related responses follow financial rules and policies.
- **Business Analyst (Sahnawag Hussain):** Acts as a link between business and technical teams by gathering requirements, clarifying expectations, and tracking chatbot performance using reporting and analytics.

#### d. Scope and Assumptions

##### In Scope

- Leave policies
- Expense reimbursement guidelines
- General HR FAQs

##### Out of Scope

- Salary changes
- Access to confidential employee performance data
- Legal or grievance handling

##### Assumptions

- All HR policies are available in updated digital formats (PDFs).
- Employees will access the chatbot using existing internal platforms.

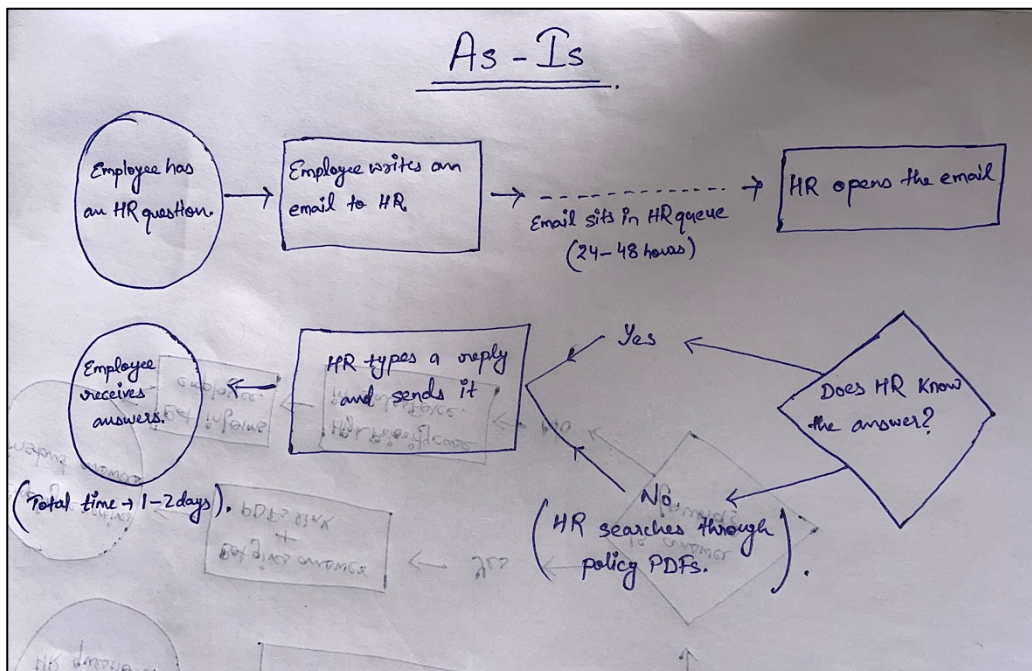
## 2. Process Analysis

A critical part of this requirement analysis was performing a thorough Process Analysis to understand how information currently flows within the HR department. By mapping the As-Is and To-Be states, I identified the specific bottlenecks that this AI solution will resolve.

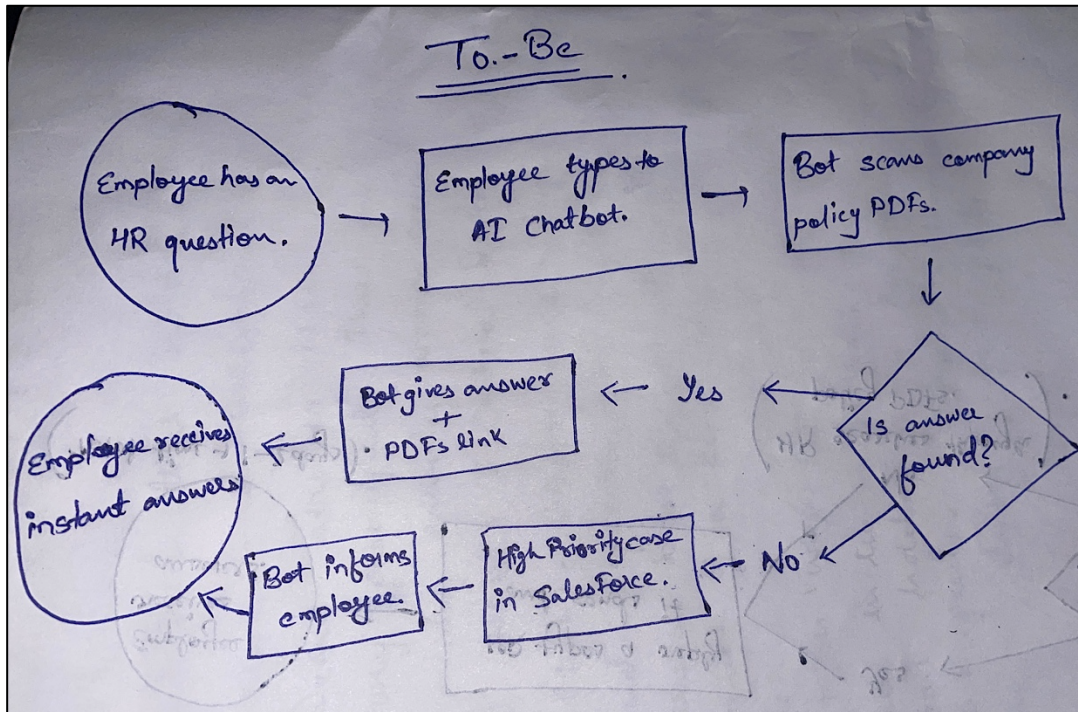
#### Understanding the Change

The following diagrams illustrate the transition from a manual, high-latency process to an automated, data-driven workflow:

- The As-Is Model: Highlights the current "Pain Points," such as the 24–48 hour delay caused by the manual HR email queue and the time spent by staff searching through folders.



- The To-Be Model: Demonstrates the "Future State" where an AI Chatbot provides instant resolution. It also introduces a "Fail-Safe" logic: if the bot cannot find an answer, it automatically triggers a High-Priority Case in Salesforce to ensure no employee is left without help.



### 3. User Stories

#### User Story 1

As an employee, I want to check my remaining leave balance, so that I can plan my time off without contacting HR.

##### Acceptance Criteria

- The chatbot fetches leave balance using the employee's ID.
- The remaining leave is displayed clearly and in simple language.

#### User Story 2

As an employee, I want to understand the reimbursement process, so that I can submit my expenses correctly the first time.

##### Acceptance Criteria

- The chatbot lists required documents such as bills or receipts.
- A direct link to the reimbursement form is provided.

#### User Story 3

As an HR Manager, I want to upload updated policy documents, so that the chatbot reflects the latest information.

### Acceptance Criteria

- Newly uploaded documents become searchable within a few minutes.
  - Older or outdated policies are marked clearly or removed from active use.
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### User Story 4

As a Business Analyst, I want visibility into unanswered chatbot queries, so that gaps in HR documentation can be identified.

### Acceptance Criteria

- Unanswered queries are stored in a database.
  - A weekly report highlights recurring unanswered questions.
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### User Story 5

As an IT administrator, I want role-based access to chatbot logs, so that sensitive employee information is only visible to authorized users.

### Acceptance Criteria

- Only users with admin roles can view detailed chat logs.
  - Sensitive fields (employee ID, payroll terms) are masked for non-authorized users.
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## 4. High-Level Solution Architecture

To ensure the chatbot is both smart and measurable, I have designed a logical structure that connects our company data with the AI engine. This architecture focuses on accuracy and data tracking.

### Tools & Technologies

- **User Interface (Frontend):** The chatbot will be integrated with platforms such as Slack, Microsoft Teams, or the internal company portal which will allow employees to get help without leaving the platforms they already use for work.
- **AI Engine:** We will use an API-based model (like Open-AI GPT-4 or Google Gemini). To prevent the AI from making up wrong answers, it will use RAG (Retrieval-Augmented Generation) logic.
- **Knowledge Base:** Central storage of verified HR policy PDFs indexed for search.
- **Data Logging (SQL):** SQL database to store chat interactions for tracking and analysis.
- **Performance Monitoring:** Tableau dashboards to monitor usage, unanswered queries, and overall performance.
- **Human Handoff (Salesforce):** If the bot cannot find a solution, it will automatically trigger a High-Priority Case in Salesforce so the HR team can step in immediately.

### The Logical Flow:

1. Employee asks a question.
2. AI Engine searches the verified PDF Knowledge Base.
3. System provides the answer plus a link to the source document for proof.
4. SQL Database logs the interaction for future analysis.
5. Salesforce receives a case only if the bot cannot resolve the issue.

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## 5. Success Metrics & Analytics (KPIs)

To check whether the HR chatbot is actually solving the problem and not just existing as a tool, the following metrics will be tracked on a regular basis. These metrics focus on efficiency, accuracy, and employee experience.

### 1. Query Resolution Rate

**What it measures:**

The percentage of employee questions that are handled fully by the chatbot without HR involvement.

**How it will be measured:**

Chat records will be stored in a SQL database. The number of completed chatbot conversations will be compared against the number of HR tickets raised for the same period.

**Target:**

Around **70% of queries** resolved by the chatbot within the first few months of launch.

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### 2. Average Response Time

**What it measures:**

How quickly employees receive useful answers compared to the earlier email-based process.

**How it will be measured:**

The time difference between when a question is asked and when a response is provided will be calculated using chat logs. This will help show how much time is saved for employees.

**Reporting:**

A simple trend view in Tableau to compare response times before and after chatbot adoption.

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### 3. Unanswered or Escalated Queries

**What it measures:**

Questions that the chatbot is unable to answer and needs to escalate to HR.

**How it will be measured:**

Queries marked as “unanswered” or escalated will be reviewed weekly to identify missing or unclear policy information.

**Action Taken:**

These insights will be shared with the HR team so policies or documents can be updated where required.

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### 4. Employee Feedback Score

**What it measures:**

Employee satisfaction with the chatbot responses.

**How it will be measured:**

After a chat session, employees can give quick feedback using a thumbs-up or thumbs-down option.

**Use of data:**

Feedback trends will be monitored to understand where responses are helpful and where improvements are needed.

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